

5.4.4

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Question

Using elementary transformations, find the inverse of the following matrix

$$\begin{pmatrix} 2 & 4 \\ -1 & 2 \end{pmatrix} \quad (1)$$

We know that

$$\mathbf{A}^{-1}\mathbf{A} = \mathbf{I} \quad (2)$$

where $\mathbf{I}$ is the 2×2 identity matrix.

So we form the augmented matrix:

$$\left(\begin{array}{cc|cc} 2 & 4 & 1 & 0 \\ -1 & 2 & 0 & 1 \end{array} \right) \quad (3)$$

Row Transformations

Applying row operations:

$$\left(\begin{array}{cc|cc} 2 & 4 & 1 & 0 \\ -1 & 2 & 0 & 1 \end{array} \right) \xleftrightarrow[R_2 \leftarrow R_2 + \frac{1}{2}R_1]{R_1 \leftarrow \frac{1}{2}R_1} \left(\begin{array}{cc|cc} 1 & 2 & 0.5 & 0 \\ 0 & 3 & 0.5 & 1 \end{array} \right) \quad (4)$$

$$\xleftrightarrow[R_1 \leftarrow R_1 - \frac{2}{3}R_2]{R_2 \leftarrow \frac{1}{3}R_2} \left(\begin{array}{cc|cc} 1 & 0 & \frac{1}{3} & -\frac{2}{3} \\ 0 & 1 & \frac{1}{6} & \frac{1}{3} \end{array} \right) \quad (5)$$

Therefore, the inverse is

$$\mathbf{A}^{-1} = \begin{pmatrix} \frac{1}{3} & -\frac{2}{3} \\ \frac{1}{6} & \frac{1}{3} \end{pmatrix} \quad (6)$$

This can be verified in code by showing

$$\mathbf{A}^{-1}\mathbf{A} = \mathbf{I}. \quad (7)$$

```
#include <stdio.h>

void matmul(double* A, double* B, double* C) {
    for (int i = 0; i < 2; i++) {
        for (int j = 0; j < 2; j++) {
            double sum = 0.0;
            for (int k = 0; k < 2; k++) {
                sum += A[i*2 + k] * B[k*2 + j];
            }
            C[i*2 + j] = sum;
        }
    }
}
```

Python + C Code

```
import ctypes
import numpy as np

lib = ctypes.CDLL("./libmatmul.so")
lib.matmul.argtypes = [
    np.ctypeslib.ndpointer(dtype=np.float64, flags="C_CONTIGUOUS"
        ),
    np.ctypeslib.ndpointer(dtype=np.float64, flags="C_CONTIGUOUS"
        ),
    np.ctypeslib.ndpointer(dtype=np.float64, flags="C_CONTIGUOUS"
        ),
]
lib.matmul.restype = None
A = np.array([[2, 4],
              [-1, 2]], dtype=np.float64)
B = np.array([[1/3, -2/3],
              [1/6, 1/3]], dtype=np.float64)

C = np.zeros((2, 2), dtype=np.float64)
```

Python Code

```
import numpy as np
a = np.array([[2, 4],
              [-1, 2]])
inverse_a = np.array([[1/3, -2/3],
                      [1/6, 1/3]])
b = a @ inverse_a
print(b)
```